

Gabriel Mello

Software Engineer · AI Integration · Full-Stack
3+ yrs JS/TS · React · Next.js · Node · Production LLM integration

📍 Jacarezinho, PR — Brazil
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🔓 Open to remote, hybrid and on-site opportunities

ABOUT

Software Engineer with 3+ years of professional experience across JavaScript/TypeScript, React, Next.js and Node.js. I work on the intersection between full-stack engineering and applied AI — building systems where LLM-powered features have to actually work in production, in front of real users, with non-technical stakeholders depending on them.

I do my best work when the problem is open-ended: ambiguous requirements, undocumented codebases, integrations between systems that were never meant to talk to each other. I get pulled in to map the situation, define the architecture, ship the solution, and keep it running.

I run my own SaaS product in production (FlunexApp), build institutional AI tooling at a Brazilian state university (UENP), and contribute to open-source / volunteer projects on the side.

WHAT I DO BEST

- Full-stack engineering — React, Next.js, TypeScript on the frontend; Node.js (Express) and Python (Django) on the backend.
- AI agent integration — designing prompting strategies, RAG pipelines and validation loops that keep LLM features reliable in production.
- Backend architecture — REST APIs, multi-tenancy, database modeling (PostgreSQL), authentication, observability.
- Onboarding into unfamiliar codebases — legacy CMSs, government platforms, integration layers; mapping them fast and shipping changes safely.
- Translating business requirements — gathering needs directly with non-technical stakeholders (legal, finance, public-sector partners) and turning them into concrete technical solutions.

FEATURED PROJECTS

FlunexApp — Multi-tenant SaaS for WhatsApp automation

Founder & Lead Engineer · In production · 2025 – Present

Multi-tenant SaaS architected and built from scratch. Visual conversational flow editor with AI-powered flow generation using multiple LLM providers (Gemini, Claude, Groq). Recurring subscription billing with proration and credits. Strict multi-tenancy, multi-workspace with role-based access control, full pt-BR / en internationalization, and a complete anti-abuse and data-protection system running in production.

- Dual-process architecture: Next.js web layer + dedicated Node.js worker.
- AI flow generation orchestrated across multiple LLM providers with response validation and fallback.
- Recurring billing integration, idempotent webhook handling, HMAC validation.
- Continuous feedback loop: agent output validation and prompt refinement.

Stack: Next.js, React, TypeScript, Node.js (Express), PostgreSQL, Redis, Docker, Tailwind CSS.

ETP Generator — Institutional AI document tool

Designer & Implementer · Live across every UENP campus · 2025 – Present

An AI-powered Python tool that generates official institutional documents (Estudo Técnico Preliminar) for the entire university. Built end-to-end: requirement gathering with legal and finance departments, RAG pipeline around real government documents, prompt design and output validation. A process that used to take hours of human rework now finishes in under ten minutes, in the exact institutional format required.

- Built and deployed across every UENP campus.
- RAG pipeline grounded in real government documents.
- Continuous refinement of agent behavior to meet institutional quality bar.

- Direct stakeholder collaboration (legal + finance) for requirement design.

Stack: Python, Django, PostgreSQL, RAG, Gemini / Claude / GPT-4.

AITEC Migration — Legacy CMS replacement

Sole engineer · Live · 2025

Replaced an outdated, undocumented Joomla CMS with a modern Next.js + Redis stack featuring multi-layer caching. Built a custom admin panel so the non-technical team could manage content autonomously.

- 10× to 20× performance gain measured in initial benchmarks.
- No server infrastructure changes required.
- Admin panel removed the team's dependency on technical support.

Stack: Next.js, React, TypeScript, Redis, ShadCN UI, TanStack Query.

TV Box AI Kiosk × Brazilian Federal Revenue

AI layer architect · In development · 2026

Partnership project between UENP and Brazil's Federal Revenue Service: a self-service kiosk based on TV Box hardware with a conversational AI agent that answers taxpayers' questions. I work on the LLM layer architecture, prompt design for a high-trust public-sector context, and the UX adapted for a constrained public terminal.

- High-trust public-sector AI deployment.
- Custom prompt design and validation strategy.
- UX adapted for constrained public-terminal usage.

Stack: Python, LLM integration (Gemini / Claude), institutional knowledge base.

Fazenda Escola — Institutional website

Full-stack developer · Live · 2025

Institutional platform for the Fazenda Escola (UENP's teaching farm). Responsive site with a custom admin panel for autonomous management of educational content, extension projects and program visibility.

Stack: Next.js, React, TypeScript, Tailwind CSS.

SUAP Integrations

Integration engineer · Live · 2025 – Present

Designed the connections between UENP's central system (SUAP — Django-based institutional platform) and external platforms running on WordPress/PHP and Next.js. Ensures data consistency on both ends and eliminates manual steps in interdepartmental administrative flows.

Stack: Python, Django, REST APIs, WordPress/PHP, Next.js.

Care4You — Clinical management platform (volunteer)

Volunteer Front-End Developer · Active · Nov 2023 – Present

Volunteer work at SouJunior Labs on a clinical management platform built collaboratively with designers and a back-end team. Real-world environment for making frontend decisions that hold up as the codebase grows.

Stack: Next.js, React, TypeScript, Tailwind CSS, Zustand, Figma.

TECHNICAL SKILLS

Frontend	React · Next.js · TypeScript (strict) · Angular · Tailwind CSS · ShadCN UI · TanStack Query · Zustand · Framer Motion
Backend	Node.js (Express) · Python (Django basics) · REST APIs · Authentication / Authorization · Webhooks (HMAC) · Background jobs
Databases & Cache	PostgreSQL · MySQL · Redis (cache + pub/sub)
AI / LLMs	Gemini · Claude · GPT-4 · Groq · RAG · Prompt Engineering · Agent behavior validation
Cloud / DevOps	AWS (EC2, RDS, VPC, S3, IAM — 5 certifications) · Docker · CI/CD · Grafana · Git /

	GitHub · Scrum / Jira
Quality	Jest · Testing Library · Zod · TypeScript strict · ESLint · Code review
Mobile	React Native

EXPERIENCE

Developer Analyst

Feb 2025 – Present

Universidade Estadual do Norte do Paraná (UENP) · Jacarezinho, PR · Hybrid

Responsible for the development and maintenance of the university's digital ecosystem, working independently on projects that affect every campus. Designed and shipped AI-powered tools running across all campuses, replaced a legacy CMS with measurable performance gains, designed system integrations and participate in architecture and code-standard decisions.

Founder & Lead Engineer

2025 – Present

FlunexApp · Independent product · Remote

Architected and built a multi-tenant SaaS from scratch. Own the full lifecycle: architecture, development, infrastructure, billing and ongoing operation.

Volunteer Front-End Developer

Nov 2023 – Present

SouJunior Labs — Care4You · Remote

Collaborative frontend work in a production-like environment alongside designers and a back-end team.

EDUCATION

Associate Degree in Systems Analysis and Development · Feb 2019 – Aug 2022

FATEC Ourinhos — Faculdade de Tecnologia de Ourinhos

Extension Program in Computer Science

Universidade Estadual do Norte do Paraná (UENP)

CERTIFICATIONS

- AWS Educate: Cloud 101 · Security · Networking · Databases · Compute & Storage — Amazon Web Services · 2025
- NLW Journey — React Native — Rocketseat · 2024
- React Zero to Mastery + Next.js + Redux — Udemy / Origamid · 2023–2024
- Front-End Bootcamp: HTML, CSS, JS, React+ — Udemy · 2023
- OutSystems Training — Tata Consultancy Services · 2023
- UI Design for Beginners — Origamid · 2024

LANGUAGES

?? Portuguese: Native / Fluent ?? English: Fluent technical reading and writing · Basic conversation

CONTACT

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